MTH207

Examination - July Semester 2024

Linear Algebra

Wednesday, 13 November 2024 04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SIX (6)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Consider the system of linear equations:

$$\begin{aligned} x + y + z &= 1 \\ 4x + 6y - z &= 0 \\ -x + \alpha y + z &= \beta \end{aligned}$$

where $\alpha, \beta \in \mathbb{R}$ are real constants. Solve this system of linear equations and determine the conditions on α and β such that the linear system has

Question 1a(i)

One unique solution.

(3 marks)

Question 1a(ii)

No solutions.

(4 marks)

Question 1a(iii)

Infinitely many solutions.

(3 marks)

Question 1b

Let A and B be square matrices of order n. Suppose they each satisfy the following equations:

$$\begin{aligned} A^4 - 5A^5 + 7I_n &= 0 \\ B^4 - 4B^3 + B^2 &= 0 \end{aligned}$$

Question 1b(i)

Is A necessarily invertible? Justify your answer.

(4 marks)

Question 1b(ii)

Is B necessarily invertible? Justify your answer.

(4 marks)

Question 2a

Let $A = \begin{pmatrix} k & 1 & 0 \\ k^2 & -1 & k \\ -3 & k & k \end{pmatrix}$, where k is a real number.

Question 2a(i)

Calculate the determinant of A and leave your answer in terms of k. Hence determine all values of k such that A is invertible.

(4 marks)

Question 2a(ii)

Consider when $k = 2$; i.e.

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 4 & -1 & 2 \\ -3 & 2 & 2 \end{pmatrix}.$$

Find a 2×2 matrix of the form

$$B = \begin{pmatrix} -3 & 2 \\ x & y \end{pmatrix}$$

such that $\det A = 2 \det B$. Justify whether your answer is unique or not.

(4 marks)

Question 2a(iii)

Suppose now A is invertible. The inverse matrix of A is known to be of the form:

$$A^{-1} = \frac{1}{F(k)} \begin{pmatrix} k+1 & 1 & -1 \\ G(k) & -k & k \\ \frac{-k^3+3}{k} & H(k) & k+1 \end{pmatrix}$$

where $F(k), G(k)$ and $H(k)$ are in terms of k. Determine the expression of $F(k), G(k)$ and $H(k)$ fully.

(8 marks)

Question 2b

Using your results from Question 2a or otherwise, solve fully the system of linear equations:

$$\begin{aligned} x + y &= 1 \\ x - y + z &= -2 \\ -3x + y + z &= 0 \end{aligned}$$

(5 marks)

Question 3a

Let
$S = \left\{ \begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix}^T, \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix}^T, \begin{pmatrix} 2 & 1 & 0 & 2 \end{pmatrix}^T, \begin{pmatrix} 0 & -2 & 3 & 3 \end{pmatrix}^T \right\}.$

Question 3a(i)

Show that set S spans the entirety of $\mathbb{R}^4$; i.e. $\mathbb{R}^4 = \operatorname{span}(S)$.

(4 marks)

Question 3a(ii)

Consider the matrix

$$A = \begin{pmatrix} 1 & 1 & 2 & 0 \\ 1 & 0 & 1 & -2 \\ 0 & -1 & 0 & 3 \\ 0 & 0 & 2 & 3 \end{pmatrix}.$$

Using suitable elementary row operations, i.e. with the help of an augmented matrix, compute A^{-1}.

(7 marks)

Question 3a(iii)

Express the vector $\mathbf{v} = \begin{pmatrix} 7 & 1 & 2 & 3 \end{pmatrix}^T$ as a linear combination of the vectors in S.

(4 marks)

Question 3a(iv)

Let $T : \mathbb{R}^4 \to \mathbb{R}^4$ be a linear transformation given by A as its standard matrix. Determine the following:

(i) the nullspace of T.

(ii) the range of T.

(iii) the linear transformation $S : \mathbb{R}^4 \to \mathbb{R}^4$ such that $S \circ T = I$, where I denotes the identity map.

(4 marks)

Question 3b

(Parallelogram law) Show that for any vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, one has

$$||\mathbf{x}+\mathbf{y}||^2 + ||\mathbf{x}-\mathbf{y}||^2 = 2\left(||\mathbf{x}||^2 + ||\mathbf{y}||^2\right).$$

(5 marks)

Question 4

The sequence of **Pell numbers,** $\{P_n\}_n$**,** is defined by:

$$P_n = 2P_{n-1} + P_{n-2}, n \geq 2$$

and $P_0 = 0$ and $P_1 = 1$.

Question 4a

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$.

Question 4a(i)

Calculate the eigenvalues of A as well as their corresponding eigenvectors.

(8 marks)

Question 4a(ii)

Hence, write down an invertible matrix P and a diagonal matrix D such that one has

$$A = PDP^{-1}.$$

(2 marks)

Question 4b

Show that

$$\begin{pmatrix} P_n \\ P_{n-1} \end{pmatrix} = A^{n-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

for all $n \geq 1$. Hence show that Pell numbers

$$P_n = a^n - b^n, n \geq 0$$

where a and b are constants to be determined.

(6 marks)

Question 5a

Let $A = \begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 & -1 \\ 2 & 0 & 1 & 0 & 1 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 \end{pmatrix}^T$.

Question 5a(i)

Verify that $\mathbf{x}$ does not lie in the nullspace of A.

(2 marks)

Question 5a(ii)

Apply the Gram-Schmidt process to determine an orthogonal basis for the nullspace of A.

Remark: You can take the RREF of A as $\begin{pmatrix} 1 & 0 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & -0.25 & 0.5 \\ 0 & 0 & 1 & -1 & 1 \end{pmatrix}$

(8 marks)

Question 5a(iii)

Find a vector $\mathbf{y} \in \mathbb{R}^5$ in the nullspace of A such that $||\mathbf{y} - \mathbf{x}||$ is at the minimum value.

(6 marks)

Question 5a(iv)

Hence, show that one can write $\mathbf{x} = \mathbf{x}_r + \mathbf{x}_n$ where $\mathbf{x}_r$ lie in the row space of A and $\mathbf{x}_n$ lie in the nullspace of A.

(5 marks)

----- END OF PAPER -----